

```
jishnu@jishnu-VirtualBox:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libaio1 libevent-core-2.1-6 libhtml-template-perl
  mysql-client-5.7 mysql-client-core-5.7 mysql-common
  mysql-server-5.7 mysql-server-core-5.7
Suggested packages:
  libipc-sharedcache-perl mailx tinyca
The following NEW packages will be installed:
  libaio1 libevent-core-2.1-6 libhtml-template-perl
  mysql-client-5.7 mysql-client-core-5.7 mysql-common
  mysql-server-5.7 mysql-server-core-5.7
0 upgraded, 9 newly installed, 0 to remove and 293 not upgraded.
Need to get 19.1 MB of archives.
```

Figure 2: Installing mysql-server

```
root@jishnu-VirtualBox:/home/jishnu# apt install bacula-server bacula-client
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bacula-bscan bacula-common bacula-common-ppsql
  bacula-console bacula-director bacula-director-ppsql
  bacula-fd bacula-sd bsd-mailx dbconfig-common
  dbconfig-ppsql libblockfile-bin libblockfile1 libpq5 mt-st
  mtr postfix postgresql postgresql-10 postgresql-client
  postgresql-client-10 postgresql-client-common
  postgresql-common sysstat
Suggested packages:
  bacula-doc gawk postgresql-contrib dds2tar scsistools
  sg3-utils procmail postfix-mysql postfix-ppsql
  postfix-lsp postfix-core postfix-lmb postfix-sqlite
  sasl2-bin dovecot-common resolvconf postfix-cdb
  postfix-doc postgresql-doc locales-all postgresql-doc-10
  libjson-perl lsg
The following NEW packages will be installed:
  bacula-bscan bacula-client bacula-common
  bacula-common-ppsql bacula-console bacula-director
  bacula-director-ppsql bacula-fd bacula-sd bacula-server
  bsd-mailx dbconfig-common dbconfig-ppsql libblockfile-bin
  libblockfile1 libpq5 mtr mtr postfix postfix-bin
  postfix-common postfix-config postfix-files postfix-script
  postgresql-10 postgresql-client postgresql-client-10
  postgresql-client-common postgresql-common sysstat
0 upgraded, 26 newly installed, 0 to remove and 293 not upgraded.
Need to get 8,684 kB of archives.
After this operation, 31.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu bionic/universe amd64 bacula-common amd64 9.0.6-1build1 [776 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 libpq5 amd64 10.17-0ubuntu0.18.04.1 [107 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu bionic/universe amd64 bacula-common-ppsql amd64 9.0.6-1build1 [13.6 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu bionic/main amd64 mtr amd64 1.3.12-10 [96.3 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu bionic/universe amd64 bacula-sd amd64 9.0.6-1build1 [124 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu bionic/universe amd64 bacula-bscan amd64 9.0.6-1build1 [22.7 kB]
```

Figure 3: Installing bacula-server and bacula-client

```
sudo apt install mysql-server
```

```
sudo apt install bacula-server bacula client
```

Then get *sudo* privileges by entering:

```
sudo su
```

Now enter the commands shown in Figure 4 and 5.

```
jishnu@jishnu-VirtualBox:~$ sudo su
[sudo] password for jishnu:
root@jishnu-VirtualBox:/home/jishnu# mysql_secure_installation
Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD PLUGIN can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD plugin?

Press y|Y for Yes, any other key for No: no
Please set the password for root here.

New password:

Re-enter new password:
By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : no
... skipping.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.
```

Figure 4: mysql secure installation -1

```
Disallow root login remotely? (Press y|Y for Yes, any other key for No) : yes
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
root@jishnu-VirtualBox:/home/jishnu#
```

Figure 5: mysql secure installation -2

```
Open bacula-dir.conf
/etc/bacula

# Default Bacula Director Configuration file

#
# The only thing that MUST be changed is to add one or more
# file or directory names in the Include directive of the
# FileSet resource.
#
# For Bacula release 9.0.6 (20 November 2017) -- ubuntu 18.04
#
# You might also want to change the default email address
# from root to your address. See the "mail" and "operator"
# directives in the Messages resource.
#
# Copyright (C) 2000-2017 Kern Sibbald
# License: BSD 2-Clause; see file LICENSE-FOSS
#

Director {
    # define myself
    Name = jishnu-VirtualBox-dir
    DIRport = 9101 # where we listen for UA connections
    QueryFile = "/etc/bacula/scripts/query.sql"
    WorkingDirectory = "/var/lib/bacula"
    PidDirectory = "/run/bacula"
    Maximum Concurrent Jobs = 20
    Password = "1cD2AYgtDSB64ukXDeN5JNacdaEiWhM0-" # Console password
    Messages = Daemon
    DirAddress = 127.0.0.1
}

JobDefs {
    Name = "DefaultJob"
    Type = Backup
    Level = Incremental
    Client = jishnu-VirtualBox-fd
    FileSet = "Full Set"
    Schedule = "WeeklyCycle"
    Storage = File1
```

Figure 6: Configuration file

During installation, you'll be asked to configure the postfix. Just proceed with this, as this is like configuring the SQL database for Bacula with DB. Now that MySQL is up and running, you need to modify the Bacula configuration files in order to specify what is to be backed up and where the backup must be stored.

Open the file using the following command:

```
gedit etc/bacula/bacula-dir.conf
```

After you enter the command, a file as shown in Figure 6 will open up.

Just modify the archive device field to the path of the files you want to back up. Then give all the required permissions to the file in order to allow Bacula to access the files. You then need to install something called *webadmin* using the following command:

```
sudo apt install webadmin
```